Sanath Coonani

3rd Year Student
Integrated Masters in Computer Science and Engineering
University of Hyderabad, Hyderabad

+91-8309841450
snathkumarcoonani@gmail.com
GitHub Profile
LinkedIn Profile
LeetCode Profile

Education

- **Integrated Masters in Computer Science and Engineering** *2023-28*
  University of Hyderabad, Hyderabad

Experience

- **Temperate Technologies – ML Intern** *June – August 2025*
  Automating Low-cost Cold-Storage Using ML & Reinforcement Learning On-Site
  - Designed an environmental suitability prediction system using **Python, Pandas, Scikit-learn** for optimal crop storage conditions.
  - Developed a **reinforcement learning–based control policy** to dynamically optimize temperature and humidity under energy constraints.
  - Analyzed system performance using **NumPy, Matplotlib, Jupyter** and validated decision stability under varying environmental conditions.
  - Deployed a real-time inference API using **Flask, REST, Docker** enabling sensor-driven automated control.

- **Freelance Web Developer** *December 2024 - January 2025*
  Developed and deployed a production website for a client Remote
  - Designed and built a responsive website using **Next.js, TailwindCSS, Node.js**, serving **500+ monthly users**.
  - Worked directly with the client to gather requirements and delivered the complete solution within **2 weeks**.
  - Improved page load speed by **40%** through asset optimization and performance tuning.
  - Handled end-to-end delivery including development, cloud deployment, and post-launch maintenance.

Personal Projects

- **Geometry Dash AI Agent: Deep Reinforcement Learning Controller**
  An autonomous vision-based agent that learns optimal jump timing using Deep Q-Networks (DQN).
  - Built a reinforcement learning environment using screen capture and automated input, processing **30+ FPS** for real-time interaction.
  - Trained a CNN-based DQN on **100K+ frames**, improving average survival time by **4.5×** compared to a random baseline.
  - Implemented experience replay (**50K buffer**), epsilon decay, and target networks, reducing training variance by **35%**.
  - Accelerated training by **3×** using frame preprocessing, grayscale downsampling, and parallel environment instances.
  - Designed a custom reward function and state representation to improve learning stability in sparse-reward environments.
  - **Technologies:** Python, PyTorch, OpenCV, Reinforcement Learning, Deep Q-Network (DQN)

- **Smart Incident Detection with YOLOv8**
  A real-time computer vision system for detecting fire, smoke, and crowd density in surveillance feeds.
  - Built a YOLOv8-based object detection pipeline for real-time incident monitoring.
  - Achieved **92% detection accuracy** with real-time inference at **30+ FPS** on test data.
  - Designed for integration with live CCTV streams and automated alert mechanisms.
  - Implemented a REST-based inference service using Flask for real-time alert generation.
  - **Technologies:** Python, OpenCV, Flask, Ultralytics YOLOv8

- **ML Algorithms from Scratch**
  A collection of machine learning algorithms implemented from scratch to understand core mathematics and model behavior.
  - Implemented Linear Regression, Logistic Regression, and Decision Trees without using high-level ML libraries.
  - Achieved $\mathbf{R^2 = 0.96}$ on Salary Prediction using Linear Regression.
  - Reached **88% classification accuracy** on Heart Disease Prediction using Logistic Regression.
  - Obtained **96% accuracy** on Iris dataset using a Decision Tree classifier.
  - Derived and coded Gradient Descent, Binary Cross-Entropy loss, Entropy, and Information Gain from first principles.
  - Built modular training and evaluation pipelines to compare model performance across datasets.
  - **Technologies:** Python, NumPy, Pandas, Matplotlib

Technical Skills

- **Languages:** Python, JavaScript, C++, SQL, postgreSQL
- **Machine Learning:** PyTorch, Scikit-learn, Deep Learning, Reinforcement Learning, Computer Vision, Model Evaluation
- **Cloud & Deployment:** AWS (EC2, S3), Azure Web Apps, Docker, REST APIs, Model Deployment
- **Frameworks & Tools:** Flask, OpenCV, NumPy, Pandas, Matplotlib, Git, Jupyter
- **Web & Backend:** Node.js, Express.js, Next.js, Firebase

Certifications

- **IBM Full Stack Software Developer Professional Certificate (Coursera)**
  Completed a 12-course specialization covering frontend, backend, cloud-native development, and deployment Jan 2025
- **Developing Applications with Django and SQL (Coursera)**
  Built backend applications using Django ORM with SQLite and PostgreSQL Jan 2025
- **Developing Back-End Apps with Node.js and Express (Coursera)**
  Implemented REST APIs, middleware, authentication, and MongoDB integration Jan 2025
- **Python for Data Science, AI & Development (Coursera)**
  Covered Python fundamentals, data analysis, and introductory machine learning Jan 2025

Hackathons

- **Technovista '25 – Campus Chaos Grid (2025):** Built an ML system for classifying power grid events using sensor graph features.
- **Smart Classroom Management Hackathon (2024):** Developed an web-based prototype for automated attendance and classroom safety monitoring.

Workshops & Technical Programs

- **Shaastra 2026, IIT Madras – Deep Learning using YOLO (2026):** Hands-on training on real-time object detection, model training, and practical computer vision workflows.
- **Shaastra 2026, IIT Madras – AI & Machine Learning Workshop (2026):** Practical exposure to supervised learning, model development pipelines, and real-world ML applications.
- **Shaastra 2026, IIT Madras – Ethical Web Hacking Workshop (2026):** Covered web security fundamentals, vulnerability assessment, and basic penetration testing techniques.